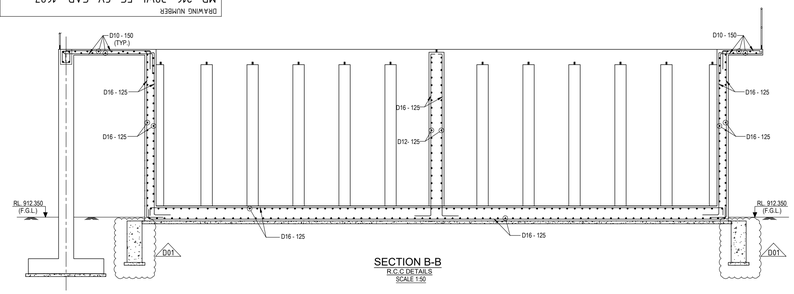
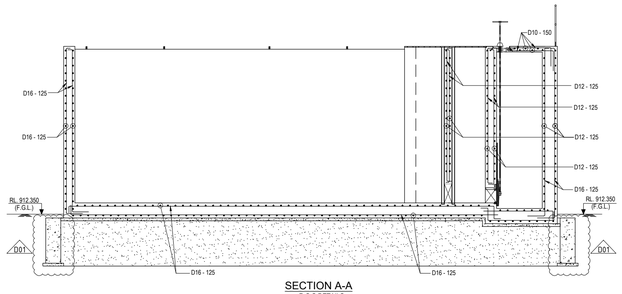
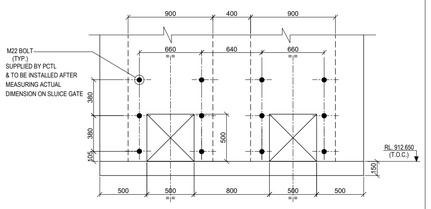
DRAWING NUMBER: MD-216-39WL-EG-CV-GAD-1697 | REV: D01



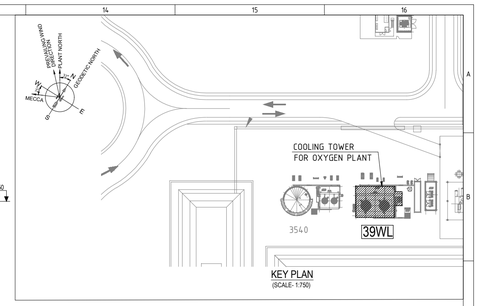
SECTION B-B
R.C.C DETAILS
SCALE 1:50

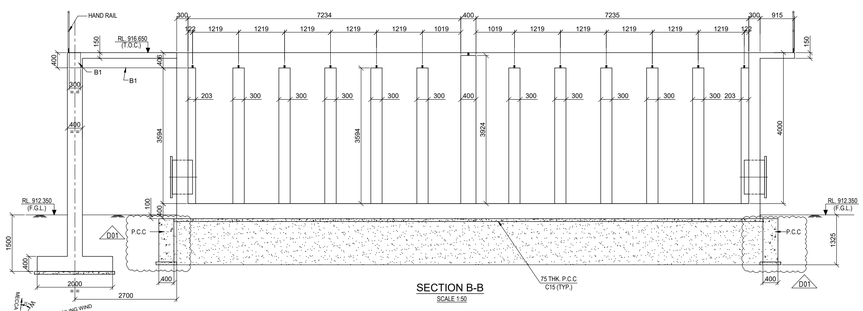


SECTION A-A
R.C.C DETAILS
SCALE 1:50

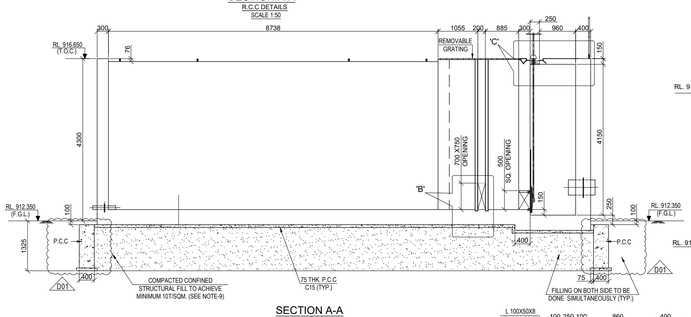


VIEW F-F
SCALE 1:20

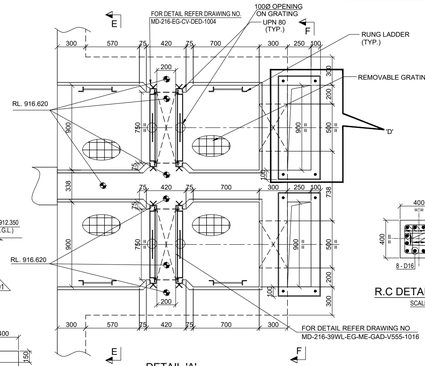




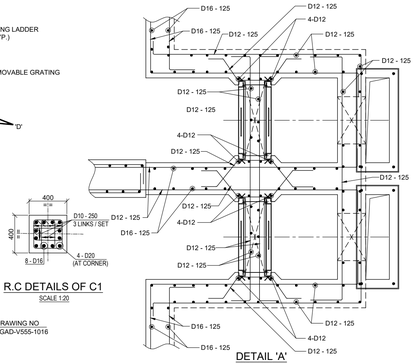
SECTION B-B
SCALE 1:50



SECTION A-A
SCALE 1:50

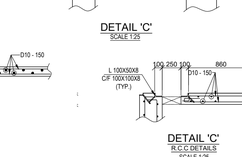


DETAIL 'A'
SCALE 1:20



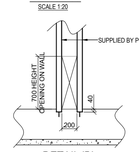
R.C DETAILS OF C1
SCALE 1:20

DETAIL 'A'
SCALE 1:20

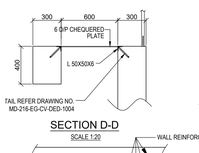


DETAIL 'C'
SCALE 1:25

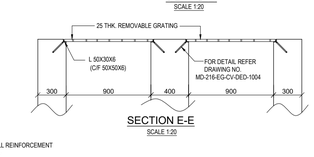
DETAIL 'C'
R.C.C DETAILS
SCALE 1:25



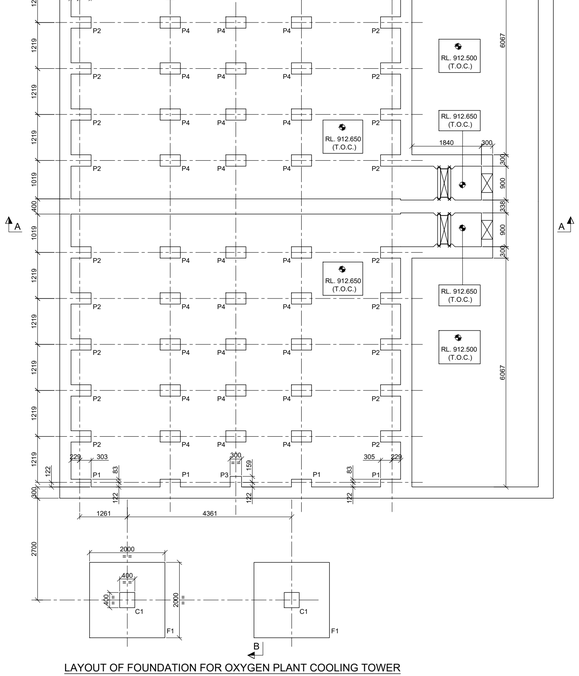
DETAIL 'B'
R.C.C DETAILS
SCALE 1:25



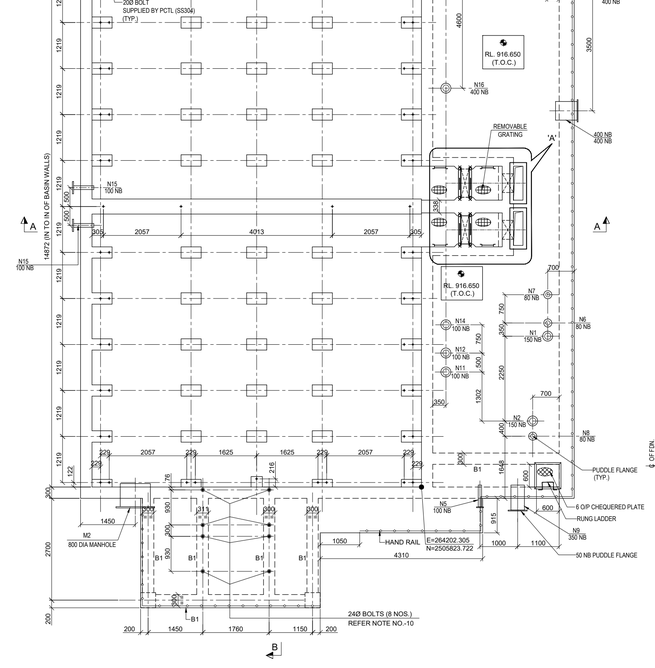
SECTION D-D
SCALE 1:20



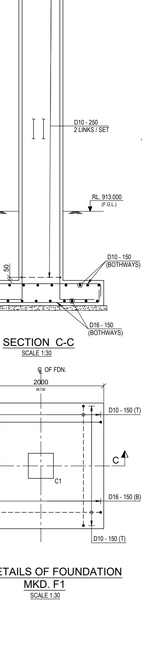
SECTION E-E
SCALE 1:20



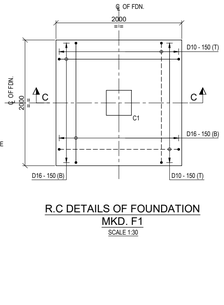
LAYOUT OF FOUNDATION FOR OXYGEN PLANT COOLING TOWER
SCALE 1:50



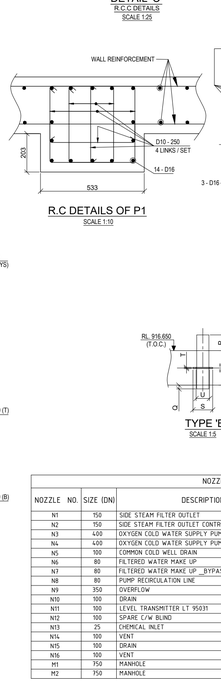
PLAN OF OXYGEN PLANT COOLING TOWER AT RL. 916.650
SCALE 1:50



SECTION C-C
SCALE 1:30

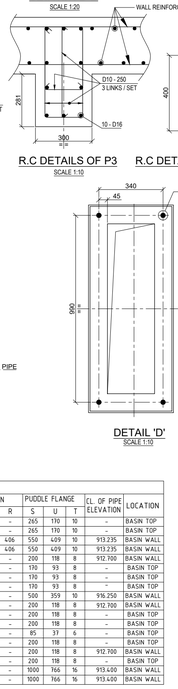


R.C DETAILS OF FOUNDATION MKD. F1
SCALE 1:30

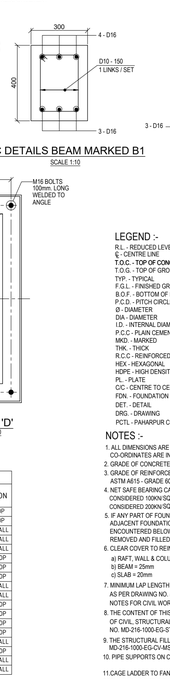


R.C DETAILS OF P1
SCALE 1:10

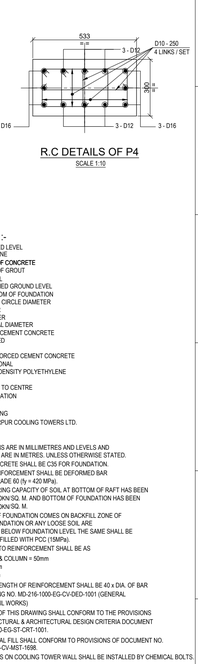
R.C DETAILS OF P2
SCALE 1:10



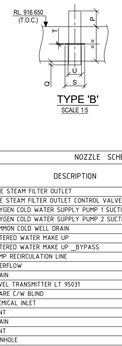
R.C DETAILS OF P3
SCALE 1:10

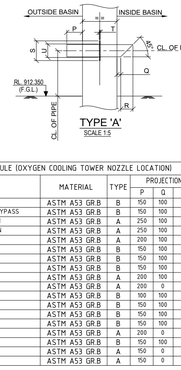


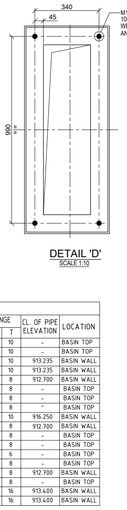
R.C DETAILS BEAM MARKED B1
SCALE 1:10



R.C DETAILS OF P4
SCALE 1:10







LEGEND :-
R.L - REDUCED LEVEL
℄ - CENTRE LINE
T.O.C - TOP OF CONCRETE
T.O.G - TOP OF GROUT
TYP. - TYPICAL
F.G.L - FINISHED GROUND LEVEL
B.O.F - BOTTOM OF FOUNDATION
P.C.D - PITCH CIRCLE DIAMETER
Ø - DIAMETER
DIA - DIAMETER
I.D. - INTERNAL DIAMETER
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED
THK. - THICK
R.C.C - REINFORCED CEMENT CONCRETE
HEX - HEXAGONAL
HDPE - HIGH DENSITY POLYETHYLENE
PL. - PLATE
C/C - CENTRE TO CENTRE
FDN. - FOUNDATION
DET. - DETAIL
DRG. - DRAWING
PCTL - PANAPUR COOLING TOWERS LTD.

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF RAFT HAS BEEN CONSIDERED 100KN/SQ. M. AND BOTTOM OF FOUNDATION HAS BEEN CONSIDERED 200KN/SQ. M.
5. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC (1:5:10MPa).
6. CLEAR COVER TO REINFORCEMENT SHALL BE AS
   a) RAFT, WALL & COLUMN = 50mm
   b) BEAM = 25mm
   c) SLAB = 20mm
7. MINIMUM LAP LENGTH OF REINFORCEMENT SHALL BE 40 x DIA. OF BAR AS PER DRAWING NO. MD-216-1000-EG-CV-DED-1001 (GENERAL NOTES FOR CIVIL WORKS)
8. THE CONTENT OF THIS DRAWING SHALL CONFORM TO THE PROVISIONS OF CIVIL, STRUCTURAL & ARCHITECTURAL DESIGN CRITERIA DOCUMENT NO. MD-216-1000-EG-ST-CRT-1001.
9. THE STRUCTURAL FILL SHALL CONFORM TO PROVISIONS OF DOCUMENT NO. MD-216-1000-EG-CV-MST-1698.
10. PIPE SUPPORTS ON COOLING TOWER WALL SHALL BE INSTALLED BY CHEMICAL BOLTS.
11. CAGE LADDER TO FAN DECK IS TO BE ANCHORED BY CHEMICAL BOLTS (SUPPLIED BY PCTL) SUIT AT SITE SITUATION AFTER INSTALLATION OF FRP COOLING TOWER.

NOZZLE SCHEDULE (OXYGEN COOLING TOWER NOZZLE LOCATION)

NOZZLE NO.	SIZE (DN)	DESCRIPTION	MATERIAL	TYPE	PROJECTION P	PROJECTION Q	PROJECTION R	PUDDLE FLANGE S	PUDDLE FLANGE U	PUDDLE FLANGE T	CL. OF PIPE ELEVATION	LOCATION
N1	150	SIDE STEAM FILTER OUTLET	ASTM A53 GR.B	B	150	100	-	265	170	10	-	BASIN TOP
N2	150	SIDE STEAM FILTER OUTLET CONTROL VALVE BYPASS	ASTM A53 GR.B	B	150	100	-	265	170	10	-	BASIN TOP
N3	400	OXYGEN COLD WATER SUPPLY PUMP 1 SUCTION	ASTM A53 GR.B	A	250	100	406	550	409	10	913.235	BASIN WALL
N4	400	OXYGEN COLD WATER SUPPLY PUMP 2 SUCTION	ASTM A53 GR.B	A	250	100	406	550	409	10	913.235	BASIN WALL
N5	100	COMMON COLD WELL DRAIN	ASTM A53 GR.B	A	200	100	-	200	118	8	912.700	BASIN WALL
N6	80	FILTERED WATER MAKE UP	ASTM A53 GR.B	B	150	100	-	170	93	8	-	BASIN TOP
N7	80	FILTERED WATER MAKE UP _BYPASS	ASTM A53 GR.B	B	150	100	-	170	93	8	-	BASIN TOP
N8	80	PUMP RECIRCULATION LINE	ASTM A53 GR.B	B	150	100	-	170	93	8	-	BASIN TOP
N9	350	OVERFLOW	ASTM A53 GR.B	A	200	100	-	500	359	10	916.250	BASIN WALL
N10	100	DRAIN	ASTM A53 GR.B	A	200	0	-	200	118	8	912.700	BASIN WALL
N11	100	LEVEL TRANSMITTER LT 95031	ASTM A53 GR.B	B	100	100	-	200	118	8	-	BASIN TOP
N12	100	SPARE C/W BLIND	ASTM A53 GR.B	B	150	100	-	200	118	8	-	BASIN TOP
N13	25	CHEMICAL INLET	ASTM A53 GR.B	B	150	100	-	85	37	6	-	BASIN TOP
N14	100	VENT	ASTM A53 GR.B	B	150	100	-	200	118	8	-	BASIN TOP
N15	100	DRAIN	ASTM A53 GR.B	A	200	0	-	200	118	8	912.700	BASIN WALL
N16	100	VENT	ASTM A53 GR.B	B	150	100	-	200	118	8	-	BASIN TOP
M1	750	MANHOLE	ASTM A53 GR.A	A	150	0	-	1000	766	16	913.400	BASIN WALL
M2	750	MANHOLE	ASTM A53 GR.A	A	150	0	-	1000	766	16	913.400	BASIN WALL

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	23/06/2023	AS - BUILT & REVISED AS MARKED	AUD	SJM	SNM	AG	
MD-216-39WL-EG-ME-GAD-V555-1021	SLUICE GATE FOR OXYGEN PLANT COOLING TOWER	C02	30/12/2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED △02	AUD	SJM	SNM	AG	
MD-216-39WL-EG-ME-GAD-V555-1013	STAIRCAE, PLATFORMS & LADDER FOR COOLING TOWER	C01	06/11/2020	ISSUED FOR CONSTRUCTION	SYA	SJM	SNM	AG	
MD-216-39WL-EG-ME-GAD-V555-1012	OXYGEN PLANT COOLING TOWER	A01	13/10/2020	ISSUED FOR APPROVAL	SYA	SJM	SNM	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_080_C_G_24976_003	3
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707__GAU01__AG11__MTL01__39198	OU100696029 / 3

DRAWN	SYA	22/09/2020
CHECKED	SJM	24/09/2020
ENGINEER	SNM	22/09/2020
APPROVED	AG	13/10/2020
EMPLOYER		
SCALE	1:50, 25, 20, 10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
OXYGEN PLANT COOLING TOWER
PLAN, ELEVATION AND DETAILS

DRAWING NUMBER: MD - 216 - 39WL - EG - CV - GAD - 1697 | REV: D01